

```
[1]: print('LAT-NUM-ROW')
      print('-----')

      for i in range(1,6):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(0,i):
              print(i,end=' ')
          print()
```

LAT-NUM-ROW

```
-----
          1
         2 2
        3 3 3
       4 4 4 4
      5 5 5 5 5
```

```
[3]: print('LAT-NUM-COL')
      print('-----')

      for i in range(1,6):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(0,i):
              print(k,end=' ')
          print()
```

LAT-NUM-COL

```
-----
          0
         0 1
        0 1 2
       0 1 2 3
      0 1 2 3 4
```

```
[4]: print('LAT-STAR')
      print('-----')

      for i in range(1,6):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(0,i):
              print('*',end=' ')
          print()
```

LAT-STAR

```

                *
              * *
            * * *
          * * * *
        * * * * *
      * * * * *
```

```
[5]: print('LAT-UPPER-ROW')
      print('-----')

      for i in range(1,6):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(0,i):
              print(chr(i+64),end=' ')
          print()
```

LAT-UPPER-ROW

```
          A
        B B
      C C C
    D D D D
  E E E E E
```

```
[6]: print('LAT-UPPER-COL')
      print('-----')

      for i in range(1,7):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(1,i):
              print((chr(k+64)),end=' ')
          print()
```

LAT-UPPER-COL

```

          A
        A B
      A B C
    A B C D
  A B C D E
```

```
[7]: print('LAT-LOWER-ROW')
      print('-----')

      for i in range(1,6):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(0,i):
              print((chr(i+64+32)),end=' ')
          print()
```

LAT-LOWER-ROW

```
          a
        b b
      c c c
    d d d d
  e e e e e
```

```
[10]: print('ILAT-NUM-ROW')
      print('-----')

      for i in range(6,0,-1):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(0,i):
              print(i,end=' ')
          print()
```

ILAT-NUM-ROW

6 6 6 6 6 6

5 5 5 5 5

4 4 4 4

3 3 3

2 2

1

```
[11]: print('ILAT-NUM-COL')
      print('-----')

      for i in range(6,0,-1):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(1,i):
              print(k,end=' ')
          print()
```

ILAT-NUM-COL

```
1 2 3 4 5
  1 2 3 4
    1 2 3
      1 2
        1
```

```
[12]: print(' ILAT-UPPER-ROW')
print('-----')

for i in range(6,0,-1):
    for j in range (6,i,-1):
        print(' ',end=' ')
    for k in range(0,i):
        print(chr(i+64),end=' ')
    print()
```

ILAT-UPPER-ROW

```
F F F F F F
  E E E E E
    D D D D
      C C C
        B B
          A
```


[13]:

```
print('ILAT-UPPER-COL')
print('-----')

for i in range(6,0,-1):
    for j in range (6,i,-1):
        print(' ',end=' ')
    for k in range(1,i):
        print((chr(k+64)),end=' ')
    print()
```

ILAT-UPPER-COL

A	B	C	D	E
	A	B	C	D
		A	B	C
			A	B
				A

```
[14]: print('ILAT-LOWER-COL')
      print('-----')

      for i in range(5,0,-1):
          for j in range (5,i,-1):
              print(' ',end=' ')
          for k in range(0,i):
              print((chr(i+64+32)),end=' ')
          print()
```

ILAT-LOWER-COL

e e e e e
 d d d d
 c c c
 b b
 a

```
[15]: print('ILAT-LOWER-COL')
      print('-----')

      for i in range(6,0,-1):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(1,i):
              print((chr(k+64+32)),end=' ')
          print()
```

ILAT-LOWER-COL

a b c d e

 a b c d

 a b c

 a b

 a

```
[16]: print('PRT-NUM-ROW')
      print('-----')

      for i in range(1,6):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(0,i):
              print(i,end=' ')
          print()
```

PRT-NUM-ROW

```
      1
     2 2
    3 3 3
   4 4 4 4
  5 5 5 5 5
```

```
print('PRT-NUM-COL')
```

```
print('-----')
```

```
for i in range(1,7):
```

```
    for j in range (7,i,-1):
```

```
        print(' ',end=' ')
```

```
    for k in range(1,i):
```

```
        print(k,end=' ')
```

```
    print()
```

```
[18]: print('PRT-UPPER-ROW')
      print('-----')

      for i in range(1,6):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(0,i):
              print((chr(i+64)),end=' ')
          print()
```

PRT-UPPER-ROW

A

B B

C C C

D D D D

E E E E E

```
[20]: print('PRT-LOWER-ROW')
print('-----')

for i in range(1,6):
    for j in range (6,i,-1):
        print(' ',end=' ')
    for k in range(0,i):
        print((chr(i+64+32)),end=' ')
    print()
```

PRT-LOWER-ROW

```
    a
  b b
c c c
d d d d
e e e e e
```

```
[21]: print('PRT-LOWER-COL')
      print('-----')

      for i in range(1,7):
          for j in range (6,i,-1):
              print(' ',end=' ')
          for k in range(1,i):
              print((chr(k+64+32)),end=' ')
          print()
```

PRT-LOWER-COL

```
      a
    a b
  a b c
a b c d
a b c d e
```